# Reader Report: Week 03

Pearl, J. (2016). *Causal Inference in Statistics: A Primer.* Ch 3–4.

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## Summary

Chapters 3 and 4 of Pearl's *Primer* formalize the distinction between *seeing* (observation) and *doing* (intervention), providing the graphical and algebraic tools to move between them.

The core contribution of Chapter 3 is the **adjustment formula** (Eq 3.5), which allows us to compute the effect of an intervention $P(Y \mid do(x))$ purely from observational probabilities $P(Y \mid x, z)$, provided we can identify a valid adjustment set $Z$. This identification is governed by the **backdoor criterion** (Def 3.3.1), which states that a set $Z$ allows for identification if it blocks all spurious paths between the treatment $X$ and outcome $Y$ without blocking directed causal paths or opening new collider paths. When the backdoor path is blocked by an unobserved confounder, the **front-door criterion** (Def 3.4.1) offers an alternative identification strategy by exploiting a mediator $M$ that intercepts the causal effect.

Chapter 4 extends this algebra to **counterfactuals**, defined as the potential outcome $Y$ had $X$ been different, given specific evidence about a **unit** $u$. Unlike interventions, which estimate the **average causal effect (ACE)** (Eq 3.1) over **populations** ($P(Y \mid do(x))$), counterfactuals **are specific to the unit** ($Y_x(u)$). They are computed via the three-step **abduction-action-prediction** algorithm: (1) *Abduct* the state of exogenous variables $U$ from **evidence** ($E = e$), (2) *Act* to **modify the structural equations** via graph surgery, and (3) *Predict* the outcome in the modified model.

## Critique & Project Connection

Pearl's graph surgery metaphor provides a rigorous definition for the invariance we will seek in Weeks 6–7. An intervention is effectively an operation that creates a new environment $e'$ where the mechanism $P(X \mid PA(X))$ is destroyed, but the mechanism $P(Y \mid PA(Y))$ remains **invariant**.

For my *Predicting NIMBYism Causally* final project for this class, this formalizes our identification strategy. We treat distinct market regimes as environments $E$. If the relationship between a feature such as school quality and an outcome such as opposition $(Y)$ changes between 2007 and 2024, it implies that the unconditional relationship is confounded by an unblocked spurious path. Our goal is to discover a feature set that effectively blocks these paths, satisfying the **backdoor criterion** across all regimes. The Counterfactual algorithm is particularly relevant for the validity check when we want to ask retrospective questions such as, "Would this specific development have been opposed if it were presented in a different year?" tailored to the specific neighborhood's exogenous characteristics $U$, such as, for example, historical wealth.

## Questions

1. When modeling multiple environments, is the environment variable $E$ simply a covariate we condition on using the adjustment formula, or does it strictly serve as a proxy for unknown intervenions?
2. Can the front-door criterion be applied to text data, where the mediator is a discovered latent topic, such as traffic congestion?